Project Proposal: Astronomical Observation Archive and Retrieval System for the INO340 Telescope

Leila Sadeghi Ardestani

Contact at: lsadeghi@ipm.ir

1. Introduction and Background

Modern astronomical observatories generate increasing volumes of observational data that require systematic storage, organization, retrieval, and long-term preservation. While telescope control systems are responsible for acquiring observations, the scientific value of these observations depends strongly on the ability to preserve associated metadata, maintain data organization, and provide efficient access to historical datasets.

The INO340 telescope currently produces observational data in the form of FITS files accompanied by observation metadata. At present, information related to observations is maintained through manually managed records, while the corresponding data files are stored separately. Although this approach can be sufficient for a limited number of observations, it becomes increasingly difficult to maintain as the number of observations increases, additional instruments are introduced, and the need for efficient scientific data retrieval grows.

Large astronomical facilities have addressed similar challenges through dedicated science archive systems. The European Southern Observatory (ESO) Science Archive provides a web-based interface for searching and retrieving astronomical observations from ESO facilities while maintaining links between observation records, raw data, calibration products, and processed datasets [1]. Similarly, the ESA XMM-Newton Science Archive (XSA) organizes observations together with their associated data products, allowing users to search datasets through scientifically meaningful parameters and access related files [2]. These archives demonstrate the importance of an observation-centric data model in which individual files are associated with higher-level observation records.

The proposed project aims to develop an Astronomical Observation Archive and Retrieval System for the INO340 Telescope, following established concepts from professional astronomical archives while adapting them to the requirements of a dedicated research observatory.

The system will provide:

- A web-based interface for searching and retrieving observations
- An automated mechanism for ingesting newly generated data
- A scalable architecture capable of supporting future instruments and additional data products.

The proposed archive shall preserve both scientific and technical metadata. Scientific metadata shall be indexed and made searchable through the user interface, enabling efficient discovery of observations. Technical metadata related to telescope and instrument characteristics shall be retained for instrument characterization, troubleshooting, calibration studies, and future analysis. This approach follows practices adopted by major observatories, where complete metadata are preserved while user-facing search interfaces focus primarily on scientifically meaningful parameters [1], [2]. The project is intended as a software engineering and computer science research project suitable for a BSc/MSc student. It combines database design, web application development, automated data ingestion, metadata management, and scalable software architecture within the context of astronomical data management.

2. Review of Existing Astronomical Archives and Design Inspiration

The design of the INO340 Astronomical Observation Archive and Retrieval System is inspired by established practices used in major astronomical archives. Although the scale of a dedicated research telescope differs from international facilities, the challenges of organizing growing data volumes, preserving metadata, and providing efficient access to observations are common.

The ESO Science Archive, XMM-Newton Science Archive (XSA), and ALMA Science Archive demonstrate several important principles relevant to the INO340 archive: observation-based organization, metadata-driven design, web-based scientific access, and long-term preservation of associated data products.

The ESO archive demonstrates the value of organizing datasets around observation records rather than individual files, while maintaining access to raw, calibration, and processed data products. The XSA approach further highlights the importance of maintaining relationships between an observation and all associated files throughout the data lifecycle. ALMA demonstrates the importance of detailed and standardized metadata for scientific discovery, interpretation, and future reuse.

Based on these principles, the INO340 archive will adopt the following design concepts:

Design Principle	Application to INO340
Observation-centric organization	Observations will be the main scientific unit, with related files and data products linked together.
Separation of data entities	Observations, data products, and physical files will be managed as separate concepts to support future expansion.

Design Principle	Application to INO340
Web-based access	Users will search and retrieve observations through a dedicated interface rather than direct database queries.
Metadata-driven design	Scientific metadata will support discovery, while technical metadata will be preserved for future analysis and instrument understanding.
Extensible architecture	Future instruments and data products will be supported without major redesign.

The INO340 archive will not attempt to reproduce the complexity of international facilities. Instead, it will adopt their fundamental design principles to create a scalable and maintainable archive suitable for a dedicated research telescope.

3. Project Objectives and Scope

The primary objective of this project is to design and develop an extensible archive and retrieval system for managing observational data from the INO340 telescope. The system shall reduce manual data management requirements while providing researchers with efficient access to structured and well-organized observational datasets.

The project focuses on automated data ingestion, metadata management, observation-centric data organization, data discovery and retrieval, and controlled user access. The archive shall operate alongside existing telescope control systems and scientific analysis software rather than replacing them. The archive shall be designed as a long-term observatory infrastructure component rather than a temporary data management solution. The system architecture shall emphasize modularity, extensibility, maintainability, and a clear separation between telescope operation and archive functions. The design shall allow future instruments, additional data products, and new scientific services to be incorporated without requiring complete redevelopment of the system. The system shall provide the following core capabilities:

3.1 Automated Data Ingestion

The system shall automatically detect newly generated FITS files from the telescope data server, extract and validate relevant metadata, associate files with the appropriate observation, and register them in the archive database without requiring manual intervention. The ingestion process shall record processing status and errors to support monitoring and troubleshooting.

3.2 Observation and Data Product Management

The archive shall organize data around observation records rather than individual files. Each observation shall be linked to its associated data products, including science frames, calibration data, raw files, and future derived products.

The architecture shall maintain a clear separation between observations, logical data products, and physical files to support future expansion and changes in data storage or processing workflows.

3.3 Web-Based Data Discovery and Retrieval

The system shall provide a web-based interface through which authorized users can search observations, inspect metadata, and retrieve associated data products.

Users shall be able to search using scientifically meaningful parameters, including target information, observation date, instrument configuration, and sky coordinates. The system shall support the retrieval of individual files as well as complete observation datasets.

3.4 Metadata Management

The system shall preserve both scientific and technical observational metadata. Scientific metadata shall be structured and indexed to support data discovery and search, while technical metadata shall be retained for instrument characterization, troubleshooting, calibration studies, and future scientific analysis.

The project shall support the development of an observatory-specific FITS metadata standard and associated metadata validation procedures where required.

3.5 Extensible Architecture

The archive shall be designed to accommodate future instruments, additional data product types, derived datasets, and new archive services without requiring complete redevelopment of the system.

The architecture shall use modular components and extensible data models to support the long-term evolution of the observatory's data management requirements.

3.6 User Management and Access Control

The archive shall implement controlled access to observational data through user roles and permissions. The system shall support different levels of access for administrators, researchers and observers, public users if public access is enabled, and automated system services.

Access control mechanisms shall determine the archive operations and datasets available to each user or service role.

3.7 System Boundaries

The archive is responsible for the ingestion, organization, metadata management, discovery, retrieval, and controlled access of observational data.

The system is not intended to replace telescope control systems, instrument control software, or scientific data analysis and reduction tools. Instead, it shall integrate with and operate alongside these systems as part of the observatory's broader data infrastructure.

3.8 Out of Scope

The project does not include telescope operation and control functions, including telescope movement, dome control, instrument operation, or observation scheduling. These responsibilities remain with existing telescope control systems.

The project will not implement observation planning functions such as proposal submission, time allocation, or scheduling of observing programs. Scientific data processing workflows, including image calibration, photometric or spectroscopic reduction, and scientific analysis pipelines, are also outside the scope of this project. However, the archive architecture should allow future integration of such capabilities.

Although technical metadata may be preserved, the archive will not replace dedicated engineering systems responsible for real-time hardware monitoring, equipment diagnostics, or instrument control.

4. System Requirements

This section defines the functional, architectural, and operational requirements of the INO340 Astronomical Observation Archive and Retrieval System. These requirements specify the expected behavior and technical characteristics of the archive within the scope defined in Section 2.

4.1 Archive Architecture and Interfaces

The archive shall operate as an independent software component integrated with the existing telescope data infrastructure.

The expected data flow shall be:

Telescope and Instrument Systems → Data Storage Server → Archive Ingestion Service → Archive Database → Web Interface

The archive shall:

- maintain structured metadata and references to archived files;
- allow FITS files to remain in their designated storage locations;
- provide controlled access to stored files through archive services; and
- avoid unnecessary duplication of observational datasets.

The architecture shall support integration with additional instruments, observing modes, and data services.

4.2 Automated Data Ingestion

The archive shall provide an automated ingestion service for registering newly generated observational data.

The ingestion service shall:

- detect newly generated FITS files;
- extract relevant FITS metadata;
- validate required metadata;
- associate files with the appropriate observations and data products;
- register archive records in the database;
- record ingestion status and errors; and
- prevent duplicate registration of previously ingested files.

The ingestion service shall be developed in coordination with the telescope control team to ensure compatibility with existing data acquisition workflows.

4.3 Metadata and Data Model Requirements

The archive shall implement an observation-centric data model that distinguishes between the following entities:

Entity	Description
Observation	The scientific representation of an observing event, including target, coordinates, observation time, instrument configuration, and access information.
Data Product	A logical scientific product associated with an observation, such as a science image, calibration dataset, or processed product.
Physical File	A stored file associated with a data product, including its location, identity, and validation information.

The archive shall maintain relationships between observations, data products, and physical files.

The data model shall allow multiple data products and physical files to be associated with a single observation without requiring changes to the core archive structure.

4.3.1 Scientific and Technical Metadata

The archive shall preserve scientific and technical metadata associated with observational data.

Scientific metadata shall be indexed where required to support archive search and discovery. Searchable parameters shall include, where applicable:

- target information;
- right ascension and declination;
- observation date and time;
- instrument configuration;
- exposure parameters; and
- filter information.

Technical metadata describing telescope and instrument state shall be preserved for instrument characterization, troubleshooting, calibration studies, and future analysis.

4.3.2 FITS Metadata Standardization

The project shall define an INO340 FITS metadata standard.

The standard shall specify:

- required FITS header keywords;
- optional FITS header keywords;
- expected metadata formats and units;
- metadata validation procedures; and
- procedures for handling missing, invalid, or inconsistent metadata.

FITS metadata generation and validation requirements shall be coordinated with the telescope control system team.

4.4 Observation and Data Product Lifecycle

The archive shall maintain the relationships between observations, data products, and physical files throughout the archival lifecycle.

The system shall:

- register newly generated observational data;
- associate science and calibration data with the appropriate observation;
- maintain data provenance relationships;
- preserve associations between logical data products and stored files; and
- support the future registration of processed and derived data products.

The initial implementation shall focus on raw observational and calibration data. Integration with scientific processing pipelines is outside the initial project scope.

4.5 Scientific Data Discovery and Retrieval

The archive shall provide a web-based interface for searching and retrieving observational data.

The search interface shall support observation-level queries using scientifically meaningful parameters, including:

- target information;
- observation date;
- instrument configuration; and
- sky coordinates using right ascension, declination, and search radius.

Users shall be able to:

- view observation metadata;
- inspect associated data products;
- retrieve individual files; and
- retrieve complete observation datasets.

Users shall not require direct database access to perform archive searches or retrieve observational data.

4.6 User Access and Archive Governance

The archive shall implement role-based access control.

The system shall support the following user categories:

- **Researchers and observers:** search and retrieve authorized observations;
- **Administrators:** manage archive operation, user permissions, and authorized metadata corrections;
- **Automated services:** register and update archive records through explicitly assigned system permissions; and
- **Public users:** access publicly released observations if public archive access is enabled.

Modification of archived metadata and data records shall be restricted to authorized users and system services.

4.7 Reliability and Data Preservation

The archive shall maintain reliable associations between database records and stored observational files.

The system shall support:

- ingestion process monitoring;
- ingestion error recording;
- detection of missing or inconsistent archive records;

- validation of references to stored files; and
- operational maintenance and recovery activities.

Backup, physical storage management, and infrastructure recovery procedures shall be coordinated with the observatory's existing data infrastructure.

4.8 Extensibility Requirements

The archive architecture shall separate core archive functionality from instrument-specific metadata and processing logic.

Core archive functions shall include:

- observation management;
- data product management;
- metadata storage;
- search and retrieval; and
- user access control.

The architecture shall allow future instruments, observing modes, and data product types to be integrated without requiring fundamental changes to the core archive system.

Future services, including scientific processing pipelines, automated quality assessment, and scientific analysis systems, are outside the initial implementation scope but shall be considered in the architectural design.

5. System Users and Access Control

The archive system shall provide controlled access to observational data through a role-based access control mechanism. Since the system will serve different types of users with different responsibilities, access privileges shall be defined according to the user's role.

The main categories of users are summarized below:

User Type	Description	Main Capabilities
Administrator	Responsible for operation, maintenance, and configuration of the archive system.	Manage users and permissions, monitor system operation, review ingestion problems, perform data re-ingestion, and manage metadata corrections.
Observer / Researcher	Primary scientific users of the archive who require access to observations and associated data products.	Search observations, view metadata, access associated files, and download data products. Users shall not modify archived metadata or alter stored observations.

User Type	Description	Main Capabilities
Public User (optional)	External users accessing observations that have been released for public use.	Search and retrieve publicly available observations according to defined access policies.
Automated Ingestion Service	A background system process responsible for transferring telescope data into the archive.	Detect new files, extract metadata, create observation records, associate files with observations, and update the archive automatically.

The archive shall support observations with different access levels, allowing the observatory to maintain private observations while providing public access to selected datasets when appropriate. Access control should be implemented at the observation level, while allowing future expansion if more detailed permissions become necessary.

The design of the access control system should support the long-term operation of the archive by allowing new user categories or modified access policies to be introduced without major changes to the underlying system.

6. Evaluation and Validation Methodology

The developed archive system shall be evaluated to verify that it satisfies the operational requirements of the INO340 observatory and the design principles defined in this proposal.

Evaluation shall focus on automated data ingestion, metadata management, the functionality of the web-based search and retrieval interface, and the ability of the architecture to support future extensions.

The system shall be tested using representative observational datasets to confirm that observations, associated files, and metadata are correctly registered, organized, and accessible.

7. Expected Outcomes and Project Deliverables

The main outcome of this project shall be a functional prototype of the INO340 Astronomical Observation Archive and Retrieval System.

The expected deliverables are:

Component	Expected Outcome
Archive database	A working database with tables/schema for observations, FITS files, metadata, instruments, and data products.

Component	Expected Outcome
Automated ingestion service	A running program/service that detects new FITS files, reads their headers, validates them, and automatically adds them to the database.
Web-based interface	A working website where users can search observations, view their details, and retrieve/download the related files.
Metadata framework	A defined list/schema of required and optional FITS metadata fields, plus rules/code for validating them.
Documentation	Technical documentation explaining how the system is designed, installed, operated, and extended.

The completed project should provide a foundation that can grow with the observatory by supporting future instruments, additional data products, and new scientific services.

8. Conclusion

The development of the INO340 Astronomical Observation Archive and Retrieval System will establish a structured framework for managing and accessing the observatory's growing volume of observational data.

By adopting proven principles from major astronomical archives while adapting them to the needs of a dedicated research telescope, the project will provide a scalable solution for automated data ingestion, metadata management, and scientific data retrieval.

The resulting system will improve the long-term usability of INO340 observations and provide a foundation for future observatory data management activities.

References

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